

Rank-three finite T-data

A computer-assisted classification

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Abstract

In the setting of identity symmetrizer and diagonal leading matrix, the classification consists of sixteen indecomposable families. Every finite-type datum is obtained from one of sixteen explicit representatives by an admissible rational rescaling of time and shifts of the three species, together with relabelling and sign exchange. The completeness argument combines a delay-independent bound on coefficient sums with exact integer feasibility checks. Two-sided reddening certificates establish periodicity of all representatives. In this setting, simultaneous positivity is both necessary and sufficient for finite type in rank three.

1 Statement of the classification

The scope is the one used in Mizuno's rank-two classification [1]: the symmetrizer is $D = I_3$, and the leading matrix is diagonal. The general definition of T-data in [2] also permits nontrivial symmetrizers and non-diagonal leading matrices.

Definition 1.1. Let r_1, r_2, r_3 be positive integers, and let $N_+(z), N_-(z)$ be 3×3 matrices over $\mathbb{N}[z]$. Set

$$N_0(z) = \text{diag}(1 + z^{r_1}, 1 + z^{r_2}, 1 + z^{r_3}), \quad A_{\pm}(z) = N_0(z) - N_{\pm}(z). \quad (1)$$

In entry (i, j) , all nonzero coefficients of $N_{\pm}$ occur at exponents $0 < p < r_i$, and the supports of the two signs are disjoint. The pair is a T-datum in the present setting if

$$A_+(z)A_-(z^{-1})^T = A_-(z)A_+(z^{-1})^T. \quad (2)$$

It is *indecomposable* if no simultaneous permutation of the indices makes both matrices block diagonal with two nonempty blocks.

Write $n_{ij;p} = [z^p](N_+ - N_-)_{ij}$ and $[a]_+ = \max(a, 0)$. The associated Y-system is

$$Y_i(u)Y_i(u - r_i) = \prod_{j=1}^3 \prod_{p=1}^{r_i-1} Y_j(u - p)^{[n_{ij;p}]_+} (1 + Y_j(u - p))^{-n_{ij;p}}. \quad (3)$$

Finite type means periodicity of this system in the universal semifield, equivalently the condition in [1, Definition 1.4].

Theorem 1.2 (Computer-assisted classification). *An indecomposable T-datum in Definition 1.1 is of finite type if and only if, after relabelling the indices and possibly exchanging the signs, it has the form*

$$A_{\pm}(z) = E(z)B_{\pm}^{(k)}(z^{\lambda})E(z)^{-1}, \quad E(z) = \text{diag}(z^{s_1}, z^{s_2}, 1), \quad (4)$$

for exactly one of the sixteen representatives in Section 7. Here $\lambda \in \mathbb{Q}_{>0}$ and $s_1, s_2 \in \mathbb{Q}$; the parameters must make every exponent integral and satisfy the support conditions in Definition 1.1.

The expression in (4) is first interpreted in the group algebra of rational powers of z . If a representative has delay $r_i^{(k)}$ and a term z^p in entry (i, j) , the transformed delay and exponent are

$$r_i = \lambda r_i^{(k)}, \quad p' = \lambda p + s_i - s_j, \quad s_3 = 0. \quad (5)$$

Thus the statement gives all polynomial lifts, including arbitrarily large delays. Admissible rescalings and shifts correspond to changes of slices; the sixteen slice classes are distinct, as discussed in Section 6.

Corollary 1.3. *An indecomposable rank-three T-datum in this setting is of finite type if and only if there is a row vector $v \in \mathbb{R}^3$ such that*

$$v > 0, \quad vA_+(1) > 0, \quad vA_-(1) > 0. \quad (6)$$

All inequalities are componentwise.

Necessity is the simultaneous positivity theorem [2, Theorem 5.5], recalled in [1, Lemma 3.3]. Sufficiency follows from the exhaustive reduction and the periodicity certificates below. Decomposable rank-three data are direct sums with rank partitions $2 + 1$ or $1 + 1 + 1$, using the rank-two and rank-one classifications.

Proof status. The proof uses exact integer and rational arithmetic and the UNSAT results of Z3. All 180 unbounded solver queries are saved and replayed. The certificate computation is reproducible; it has not been formalized in a proof assistant. The source data and runtime details are given in Appendix A.

2 Periods and the structure of the proof

Table 1 records the reddening lengths h_+, h_- in the displayed time coordinates, the labelled seed period Ω , and the number of vertices in one slice. The period is the first return of the principal C-matrix to the identity in the exact calculation.

Class	Delays (r_1, r_2, r_3)	h_+	h_-	Period	Slice size
1	(2, 2, 2)	2	4	12	3
2	(2, 2, 2)	2	7	9	6
3	(3, 3, 2)	3	8	11	8
4	(6, 2, 6)	6	14	20	7
5	(2, 2, 8)	12	12	24	6
6	(2, 2, 12)	12	24	36	8
7	(5, 2, 2)	5	7	12	9
8	(8, 2, 6)	8	14	22	8
9	(5, 2, 3)	5	8	13	10
10	(8, 2, 6)	8	16	24	8
11	(2, 2, 8)	10	8	18	6
12	(2, 2, 12)	14	18	32	8
13	(3, 2, 2)	3	7	10	7
14	(2, 12, 10)	22	18	40	12
15	(2, 2, 2)	4	3	7	6
16	(2, 2, 2)	4	3	14	6

Table 1: Periods of the sixteen representatives.

The proof has three parts. First, positivity bounds each off-diagonal coefficient sum by 7, reducing the problem to 180 constant-matrix pairs up to relabelling and sign exchange. Second, exact linear integer constraints exclude 164 pairs and classify every lift of the remaining 16 pairs. Third, positive and negative reddening sequences prove finite type for each representative, and rescaling and shifting transfer periodicity to its entire family.

3 The reduction to finitely many constant pairs

Throughout this section write $P = N_+(1)$ and $M = N_-(1)$. If the datum is of finite type, (6) gives

$$v(2I - P) > 0, \quad v(2I - M) > 0. \quad (7)$$

The diagonal entries of $2I - P$ and $2I - M$ are positive integers at most 2. Hence $P_{ii}, M_{ii} \in \{0, 1\}$. Both Z -matrices are nonsingular M -matrices, so all their principal minors are positive; this also follows by diagonal scaling using v .

Lemma 3.1 (Strong connectivity). *For every indecomposable datum in Definition 1.1, the directed graph with an arrow $i \rightarrow j$ when $i \neq j$ and $P_{ij} + M_{ij} > 0$ is strongly connected. This assertion does not require finite type.*

Proof. Suppose a cut makes both A_+ and A_- block lower triangular. Their constant terms are identity matrices. Thus $K = A_+^{-1}A_-$ exists over $\mathbb{Q}(z)$ and is regular at zero. The symplectic identity implies

$$K(z) = K(z^{-1})^\top.$$

The block lower triangular matrix K is therefore block diagonal.

If a crossing block were nonzero, let $h > 0$ be the smallest degree of an entry crossing the cut. The identity

$$K = I + A_+^{-1}(N_+ - N_-)$$

shows that the degree- h crossing coefficient of K equals the degree- h crossing coefficient of $N_+ - N_-$. A term involving a nonconstant factor of A_+^{-1} has larger degree, since a crossing factor already has degree at least h . Disjoint support makes this coefficient nonzero, a contradiction. A reducible support graph consequently has no edges between its strongly connected components, and the datum is decomposable. $\square$

Lemma 3.2 (Coefficient bound). *For an indecomposable finite-type datum, set $w_{ij} = \max(P_{ij}, M_{ij})$ for $i \neq j$. Then*

$$w_{ij}w_{ji} < 4, \tag{8}$$

$$w_{12}w_{23}w_{31} < 8, \quad w_{13}w_{32}w_{21} < 8. \tag{9}$$

In particular, $0 \leq P_{ij}, M_{ij} \leq 7$ for $i \neq j$.

Proof. For every nonzero w_{ij} , (7) gives $w_{ij}v_i < 2v_j$. Each arrow in a strongly connected three-vertex graph belongs to a simple directed cycle of length two or three. Multiplication along that cycle proves the displayed inequalities. The other factors in a nonzero cycle product are positive integers, giving the bound 7. $\square$

The program `enumerate_constants.py` exhausts the resulting finite set. It enumerates the six maxima w_{ij} , every ordered split (P_{ij}, M_{ij}) with that maximum, and all 64 choices of the diagonal entries. It imposes

$$(2I - P)(2I - M)^\top = (2I - M)(2I - P)^\top,$$

positive principal minors and exact simultaneous positivity. The counts are shown in Table 2. The bounds restrict coefficient sums; no upper bound is imposed on the delays.

Stage	Count
Strongly connected maxima satisfying the cycle inequalities	1,134
Ordered off-diagonal splits	1,474,290
Labelled pairs after symplectic and principal-minor tests	1,835
Orbits under relabelling and sign exchange	180
Orbits satisfying simultaneous positivity	180

Table 2: The exhaustive constant-matrix reduction.

For the positivity test, the program considers the cone given by the nine weak inequalities $v \geq 0$, $v(2I - P) \geq 0$ and $v(2I - M) \geq 0$. It is pointed because it lies in the nonnegative orthant. In dimension three its extreme rays arise from cross products of two constraint normals. Summing the admissible rays and testing the sum against the strict inequalities is an exact feasibility test. An explicit positive integer vector is saved for every retained pair.

4 All polynomial lifts, with unbounded delays

For each of the 180 constant pairs, introduce an integer variable for every delay and for each unit of coefficient mass in every entry. Units of the same sign may have equal exponents, so coefficients greater than one are included. Exponents of opposite signs in the same entry must differ. Ordering exponents within each sign and entry removes duplicate labellings.

For a matrix F over $\mathbb{Q}(z)$, write $F^*(z) = F(z^{-1})^T$. Each entry of $A_+A_-^* - A_-A_+^*$ expands as a difference of two finite multisets of Laurent monomials. Each exponent is an integral linear form in the delay and exponent variables. The polynomial identity is exactly equality of these exponent multisets.

The solver equates multiplicities at every exponent appearing on the positive side. Equality of the total multiplicities follows from the constant-matrix identity. This encoding is exact even when monomials coalesce. The resulting formulas belong to linear integer arithmetic with Boolean combinations of equalities, with no delay or exponent cutoff.

Exact computation 4.1. Of the 180 constant pairs, 164 have no admissible polynomial lift. The remaining 16 are the pairs in Section 7. For each survivor, all admissible lifts lie in one three-dimensional rational linear subspace generated by its representative's exponent vector and the two relative species shifts in (4).

For each survivor, `classify_lift_spaces.py` pairs equal exponents in a satisfying solution. This matching gives a homogeneous rational linear subspace on which the entire Laurent identity holds. The program then asks whether an admissible solution exists outside that subspace. All sixteen answers are UNSAT, and each subspace has dimension exactly three. Its independent generators are the representative's exponent vector and the two shifts s_1, s_2 .

The verifier separately reduces every exponent modulo the row-reduced linear equations and checks equality of the resulting monomial multisets. Thus the symplectic identities hold symbolically throughout the families. It also replays all 164 exclusion queries and all 16 queries for a solution outside the family. The complete SMT-LIB2 formulas and their SHA-256 hashes are saved. Every replayed answer is UNSAT.

This proves necessity in Theorem 1.2: finite type implies one of 180 constant pairs, and every admissible symplectic lift of such a pair belongs to one of the sixteen families.

5 Periodicity of the representatives and their families

For a delay tuple $\mathbf{r}$, the reconstructed exchange matrix has vertex set

$$R = \{(i, p) : 1 \leq i \leq 3, 0 \leq p < r_i\}.$$

It is constructed from [1, equation (2.2)]. A time step mutates the three vertices $(i, 0)$ and then relabels $(i, p) \mapsto (i, p - 1 \bmod r_i)$. The implementation checks skew-symmetry, commutativity of these mutations, return of the exchange matrix after every step, and inverse-step consistency.

Exact computation 5.1. Starting from the identity principal C-matrix, every representative reaches a negative permutation matrix in each time direction, at the lengths h_+ and h_- in Table 1. Exact integer mutation also gives the identity returns recorded in that table.

The two-sided reddening criterion [1, Proposition 2.7] proves finite type for each representative in every semifield. The five examples in the earlier rank-two table and the sixth rank-two tadpole case reproduce the reddening lengths of the rank-two classification as controls.

Lemma 5.2. *Finite type is invariant under the admissible transformations in (4).*

Proof. For integral species shifts, the transformation is the bijection of solutions $Y'_i(u) = Y_i(u - s_i)$. Replacing z by z^m , for a positive integer m , splits the system into m independent residue classes of time. It therefore preserves finite type in both directions.

For rational λ, s_1, s_2 , choose a positive integer L clearing their denominators. The transformed datum at z^L is an integral species shift of the representative at $z^{L\lambda}$. Apply the integral argument to both sides. $\square$

The exact reddening certificates and Lemma 5.2 prove sufficiency in Theorem 1.2 for every admissible parameter value. Together with the constant-matrix reduction they also prove Corollary 1.3.

6 Change of slices and distinctness

The sixteen constant pairs are distinct under relabelling and sign exchange. Specialization at $z = 1$ is invariant under (4), so its sixteen families do not overlap.

For the slice equivalence of [1, Section 3.1], integral time dilations repeat the same slice sequence. Species shifts change the placement of mutation events and the initial cut. They are connected by a straight path of admissible rational shifts, since the support inequalities define a convex domain. The event order changes only at coincident event times. Clearing denominators makes such a coincidence a simultaneous mutation in a valid T-datum, so the relevant exchange entry is zero and the mutations commute. Moving the initial cut rotates the cyclic sequence. Thus admissible rational rescalings and shifts give changes of slices.

Distinctness is checked directly on the cyclic mutation sequences. For each representative, `slice_equivalence.py` follows one connected component through the cyclic permutation of components. The returning loop has three mutation events and a terminal vertex permutation. Each event roots one orbit of that permutation, and these three orbits cover the vertices. Ordering each orbit from its root gives a canonical vertex labelling. The three orbit lengths and the exchange matrix in this labelling encode the loop completely.

The program exhausts cyclic rotations and adjacent commuting mutations until closure. For a loop encoded by $(B, (a, b, c), \pi)$, rotation gives

$$(\mu_a B, (b, c, \pi^{-1}a), \pi).$$

Adjacent events can be exchanged when their exchange entry at that point vanishes. Sign exchange is $B \mapsto -B$, with the word and permutation unchanged. The sixteen resulting sets of canonical encodings are disjoint. The full minimal encoding, as well as its hash, is saved for every class.

7 The sixteen representatives

For each class, the constant matrices displayed below are $A_{\pm}(1) = B_{\pm}^{(k)}(1) = 2I_3 - N_{\pm}^{(k)}(1)$. They are unchanged by the admissible rescalings and species shifts in (4); relabelling permutes their rows and columns, and sign exchange interchanges the pair.

Classes 1–4

In each class, $B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, 1 + z^{r_2}, 1 + z^{r_3}) - N_{\pm}^{(k)}(z)$.

Class 1

$$\mathbf{r} = (2, 2, 2)$$

$$N_+^{(1)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$N_-^{(1)}(z) = \begin{pmatrix} 0 & 0 & z \\ 0 & 0 & z \\ z & z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

$h_+ = 2, \quad h_- = 4, \quad \Omega = 12.$ Vertices per slice: 3.

Class 2

$$\mathbf{r} = (2, 2, 2)$$

$$N_+^{(2)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$N_-^{(2)}(z) = \begin{pmatrix} 0 & 0 & z \\ 0 & z & z \\ z & z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 1 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

$h_+ = 2, \quad h_- = 7, \quad \Omega = 9.$ Vertices per slice: 6.

Class 3

$$\mathbf{r} = (3, 3, 2)$$

$$N_+^{(3)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & z \end{pmatrix}$$

$$N_-^{(3)}(z) = \begin{pmatrix} 0 & z & 0 \\ z^2 & 0 & z^2 + z \\ 0 & z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -1 & 2 \end{pmatrix}$$

$h_+ = 3, \quad h_- = 8, \quad \Omega = 11.$ Vertices per slice: 8.

Class 4

$$\mathbf{r} = (6, 2, 6)$$

$$N_+^{(4)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & z^3 & 0 \end{pmatrix}$$

$$N_-^{(4)}(z) = \begin{pmatrix} 0 & 0 & z \\ 0 & 0 & z \\ z^5 & z^5 + z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & -1 \\ -1 & -2 & 2 \end{pmatrix}$$

$h_+ = 6, \quad h_- = 14, \quad \Omega = 20.$ Vertices per slice: 7.

Classes 5–8

In each class, $B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, 1 + z^{r_2}, 1 + z^{r_3}) - N_{\pm}^{(k)}(z)$.

Class 5

$$\mathbf{r} = (2, 2, 8)$$

$$N_+^{(5)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ z^4 & z^5 + z^3 & z^4 \end{pmatrix}$$

$$N_-^{(5)}(z) = \begin{pmatrix} 0 & z & 0 \\ z & 0 & z \\ 0 & z^7 + z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ -1 & -2 & 1 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -2 & 2 \end{pmatrix}$$

$h_+ = 12$, $h_- = 12$, $\Omega = 24$. Vertices per slice: 6.

Class 6

$$\mathbf{r} = (2, 2, 12)$$

$$N_+^{(6)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ z^8 + z^4 & z^9 + z^3 & 0 \end{pmatrix}$$

$$N_-^{(6)}(z) = \begin{pmatrix} 0 & z & 0 \\ z & 0 & z \\ z^6 & z^{11} + z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ -1 & -2 & 2 \end{pmatrix}$$

$h_+ = 12$, $h_- = 24$, $\Omega = 36$. Vertices per slice: 8.

Class 7

$$\mathbf{r} = (5, 2, 2)$$

$$N_+^{(7)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & z \\ 0 & z & z \end{pmatrix}$$

$$N_-^{(7)}(z) = \begin{pmatrix} 0 & z^4 + z & z^3 + z^2 \\ z & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -2 & -2 \\ -1 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$h_+ = 5$, $h_- = 7$, $\Omega = 12$. Vertices per slice: 9.

Class 8

$$\mathbf{r} = (8, 2, 6)$$

$$N_+^{(8)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & z \\ 0 & z^5 + z & 0 \end{pmatrix}$$

$$N_-^{(8)}(z) = \begin{pmatrix} 0 & z^2 & z^3 + z \\ 0 & 0 & 0 \\ z^5 & z^3 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & -2 \\ 0 & 2 & 0 \\ -1 & -1 & 2 \end{pmatrix}$$

$h_+ = 8$, $h_- = 14$, $\Omega = 22$. Vertices per slice: 8.

Classes 9–12

In each class, $B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, 1 + z^{r_2}, 1 + z^{r_3}) - N_{\pm}^{(k)}(z)$.

Class 9

$$\mathbf{r} = (5, 2, 3)$$

$$N_+^{(9)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & z \\ 0 & z^2 + z & 0 \end{pmatrix}$$

$$N_-^{(9)}(z) = \begin{pmatrix} 0 & z^2 & z^3 + z \\ 0 & z & 0 \\ z^2 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & -2 \\ 0 & 1 & 0 \\ -1 & 0 & 2 \end{pmatrix}$$

$h_+ = 5, \quad h_- = 8, \quad \Omega = 13.$ Vertices per slice: 10.

Class 10

$$\mathbf{r} = (8, 2, 6)$$

$$N_+^{(10)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & z \\ 0 & z^5 + z & 0 \end{pmatrix}$$

$$N_-^{(10)}(z) = \begin{pmatrix} 0 & z^7 + z & z^6 + z^2 \\ z & 0 & 0 \\ 0 & z^3 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -2 & -2 \\ -1 & 2 & 0 \\ 0 & -1 & 2 \end{pmatrix}$$

$h_+ = 8, \quad h_- = 16, \quad \Omega = 24.$ Vertices per slice: 8.

Class 11

$$\mathbf{r} = (2, 2, 8)$$

$$N_+^{(11)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & z \\ z^6 + z^2 & z^7 + z & 0 \end{pmatrix}$$

$$N_-^{(11)}(z) = \begin{pmatrix} 0 & z & 0 \\ z & 0 & 0 \\ z^4 & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ -2 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}$$

$h_+ = 10, \quad h_- = 8, \quad \Omega = 18.$ Vertices per slice: 6.

Class 12

$$\mathbf{r} = (2, 2, 12)$$

$$N_+^{(12)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & z \\ z^{10} + z^6 + z^2 & z^{11} + z & 0 \end{pmatrix}$$

$$N_-^{(12)}(z) = \begin{pmatrix} 0 & z & 0 \\ z & 0 & 0 \\ z^8 + z^4 & 0 & z^6 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & -1 \\ -3 & -2 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

$h_+ = 14, \quad h_- = 18, \quad \Omega = 32.$ Vertices per slice: 8.

Classes 13–16

In each class, $B_{\pm}^{(k)}(z) = \text{diag}(1 + z^{r_1}, 1 + z^{r_2}, 1 + z^{r_3}) - N_{\pm}^{(k)}(z)$.

Class 13

$$\mathbf{r} = (3, 2, 2)$$

$$N_+^{(13)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & z & 0 \\ 0 & 0 & z \end{pmatrix}$$

$$N_-^{(13)}(z) = \begin{pmatrix} 0 & 0 & z^2 + z \\ 0 & 0 & z \\ z & z & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -2 \\ 0 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

$h_+ = 3, \quad h_- = 7, \quad \Omega = 10.$ Vertices per slice: 7.

Class 14

$$\mathbf{r} = (2, 12, 10)$$

$$N_+^{(14)}(z) = \begin{pmatrix} 0 & 0 & 0 \\ z^2 & 0 & z^3 + z \\ z^7 + z^3 & z^9 & 0 \end{pmatrix}$$

$$N_-^{(14)}(z) = \begin{pmatrix} 0 & 0 & z \\ 0 & z^6 & 0 \\ z^9 + z^5 + z & 0 & 0 \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & 0 \\ -1 & 2 & -2 \\ -2 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 1 & 0 \\ -3 & 0 & 2 \end{pmatrix}$$

$h_+ = 22, \quad h_- = 18, \quad \Omega = 40.$ Vertices per slice: 12.

Class 15

$$\mathbf{r} = (2, 2, 2)$$

$$N_+^{(15)}(z) = \begin{pmatrix} 0 & 0 & z \\ 0 & 0 & z \\ z & z & 0 \end{pmatrix}$$

$$N_-^{(15)}(z) = \begin{pmatrix} 0 & z & 0 \\ z & 0 & 0 \\ 0 & 0 & z \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$h_+ = 4, \quad h_- = 3, \quad \Omega = 7.$ Vertices per slice: 6.

Class 16

$$\mathbf{r} = (2, 2, 2)$$

$$N_+^{(16)}(z) = \begin{pmatrix} 0 & 0 & z \\ 0 & 0 & z \\ z & z & 0 \end{pmatrix}$$

$$N_-^{(16)}(z) = \begin{pmatrix} z & 0 & 0 \\ 0 & z & 0 \\ 0 & 0 & z \end{pmatrix}$$

$$A_+(1) = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

$$A_-(1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$h_+ = 4, \quad h_- = 3, \quad \Omega = 14.$ Vertices per slice: 6.

A Reproduction and certificate files

The classification programs and certificates are under `research/rank3/`. The checked runtime is Python 3.10.6, SymPy 1.14.0, NumPy 2.2.6 and Z3 5.1.0. Z3 is installed locally under the ignored `_deps/` directory.

Run the following commands from the repository root:

```
python -m pip install --target research/rank3/_deps
z3-solver==5.1.0.0 (on the same command line)
python research/rank3/classify_rank3.py
python research/rank3/enumerate_constants.py
python research/rank3/solve_lifts.py --timeout 2000
--optimize --dynamics (on the same command line)
python research/rank3/classify_lift_spaces.py
python research/rank3/verify_classification.py
python research/rank3/slice_equivalence.py
```

A timeout is reported as UNKNOWN and cannot pass the verifier as an exclusion. The successful run has no UNKNOWN results. Table 3 describes the certificate files.

File	Content
<code>constant_candidates.json</code>	All 180 constant pairs and positive vectors.
<code>lift_feasibility.jsonl</code>	Solver outcomes, sixteen witnesses and periodicity certificates.
<code>lift_spaces.json</code>	Exact row-reduced equations of all sixteen families.
<code>verification.json</code>	The replayed UNSAT answers, query hashes and run-time versions.
<code>smt_queries/</code>	All 180 unbounded SMT-LIB2 formulas.
<code>slice_signatures.json</code>	Complete encodings distinguishing the slice classes.

Table 3: Certificate files under `research/rank3/`.

The earlier paper [2, Table 3] gives seven rank-three examples. In their original order, they correspond to classes 1, 16, 13, 10, 2, 4, 7 here. The other nine classes are additional to that table.

Building this document. The overview and matrix catalogue are generated directly from the verified JSON by `make_pdf_tables.py`. The source is `rank3-classification.tex`; the PDF is built with LuaLaTeX and `latexmk`. All matrix coefficients in the catalogue are generated from the exact witnesses.

References

- [1] Y. Mizuno, *Periodic Y-systems and Nahm sums: the rank 2 case*, SIGMA **21** (2025), 094, 17 pp. doi:10.3842/SIGMA.2025.094. arXiv:2301.13239v2.
- [2] Y. Mizuno, *Difference equations arising from cluster algebras*, arXiv:1912.05710v2 (2020).